STOCHASTIC PROCESSES · FALL 2026

Review of Probability

& Introduction to Stochastic Processes



Week 01 · Version 2.0

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Claude Opus5.5 has formatted this presentation.

Outline of Week 01

1

History & Philosophy

From gamblers' disputes to Kolmogorov's axioms

2

Random Variables

σ-fields, probability spaces, RVs as mappings

3

Density & Distribution Functions

PMF, CDF, PDF, common families, expectation

4

Joint & Conditional Distributions

Marginals, Bayes' rule, independence, Jacobians

5

Linear Correlation

Covariance, correlation coefficient, moments

6

Important Theorems

CLT, laws of large numbers, classic inequalities

7

Introduction to Stochastic Processes

Random functions of time and first examples

01

History & Philosophy

Where probability came from, and the two ways of thinking about it.

1 2 3 4 5 6 7

Born from a Gamblers' Dispute

- Probability started as a tool for **analyzing games of chance**
- **1654:** the Chevalier de Méré brings a gambling puzzle to Blaise **Pascal**; Pascal and Pierre de **Fermat** solve it in their letters
- The “double six” problem: is it a good bet that **at least one double six** appears in **24 throws** of two dice?
- **1657:** the first book, by Christiaan **Huygens**, *De Ratiociniis in Ludo Aleae* (“On Reasoning in Games of Chance”), in Latin

SOLVING DE MÉRÉ'S PROBLEM

$$P(\text{at least one double six in } n \text{ throws}) = 1 - \left(\tfrac{35}{36}\right)^n$$

$$n = 24: \ 0.491 \qquad n = 25: \ 0.506$$

24 throws is slightly unfavorable; 25 is the break-even point.

From Games of Chance to a Science



1654 — Pascal & Fermat
Letters on games of chance

1657 — C. Huygens
First book on probability

1713 — J. Bernoulli
Ars Conjectandi: first law of large numbers

1718 — A. De Moivre
The Doctrine of Chances; normal approximation (1733)

1812 — P. Laplace
Théorie Analytique des Probabilités

1933 — A. Kolmogorov
Axiomatic foundations

A RENAISSANCE

Laplace (1749–1827) carried the mathematics of games over to scientific and practical problems. Bernoulli (1654–1705) and De Moivre (1667–1754) drove the rapid development of the 18th century.

The 19th and 20th Centuries

19TH CENTURY

- Theory of errors (Gauss: least squares)
- Actuarial mathematics
- Statistical mechanics (Maxwell, Boltzmann)

20TH CENTURY

- Kolmogorov's **axiomatic approach**
- A. N. Kolmogorov, *Grundbegriffe der Wahrscheinlichkeitsrechnung*, 1933
- English: *Foundations of the Theory of Probability*, Chelsea, New York, 1950

TODAY

- Probability theory is part of **measure theory**
- Russian school: Chebyshev, Markov, Lyapunov, Kolmogorov

Two Philosophies: Frequentist vs. Bayesian

FREQUENTIST

- Parameter θ is **fixed but unknown**
- Observations X come from an underlying distribution f(X|θ) and are **sufficient** to learn it
- All inference uses the **likelihood** f(X|θ)
- For a Gaussian likelihood, the most likely mean is the sample average:

$$\hat{\mu}_{\mathrm{ML}} = \arg\max_{\mu} f(X \mid \mu) = \frac{1}{N}\sum_{i=1}^{N} x_i$$

BAYESIAN

- Parameter θ is **random but unknown**
- Observations are **necessary but not sufficient**: prior knowledge π(θ) is essential
- Combine prior and likelihood into the **posterior** f(θ|X)
- The posterior mean E[θ|X] is a natural point estimate

$$f(\theta \mid X) = \frac{f(X \mid \theta)\,\pi(\theta)}{\int f(X \mid \theta)\,\pi(\theta)\,d\theta}$$

Example: Is the Coin Biased?

A coin is tossed **1000** times: **800 heads**, 200 tails. Let p = P(heads). What is p?

FREQUENTIST ANALYSIS

$$\hat{p} = \frac{800}{1000} = 0.8, \qquad E[\hat{p}] = p$$

$$\hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{N}} = [0.775,\ 0.825]$$

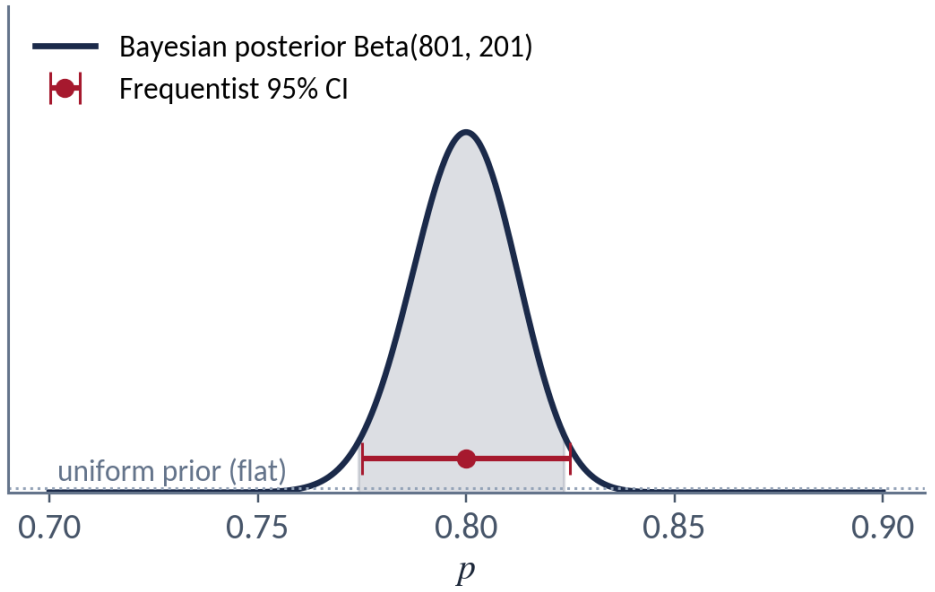
95% of intervals built this way cover the true p.

BAYESIAN ANALYSIS

$$\pi(p) = 1, \quad 0 \le p \le 1 \quad (\text{Uniform}(0,1))$$

$$\pi(p \mid \text{data}) \propto p^{800}(1-p)^{200} \ \Rightarrow\ p \mid \text{data} \sim \text{Beta}(801, 201)$$

$$E[p \mid \text{data}] = \frac{801}{1002} \approx 0.799, \qquad P(0.774 \le p \le 0.824 \mid \text{data}) = 0.95$$



Further Reading

TEXTBOOKS

- A. Papoulis, S. U. Pillai, *Probability, Random Variables and Stochastic Processes*, 4th ed., McGraw-Hill, 2002
- A. Leon-Garcia, *Probability, Statistics, and Random Processes for Electrical Engineering*, 3rd ed., Pearson, 2008

HISTORY OF PROBABILITY

- http://www.leidenuniv.nl/fsw/verduin/stathist/stathist.htm
- http://www.mrs.umn.edu/~sungurea/introstat/history/indexhistory.shtml
- https://www.cs.ucl.ac.uk/staff/D.Wischik/Talks/histprob.pdf

02

Random Variables

Sets, σ-fields and probability spaces: the ground a random variable stands on.

1 2 3 4 5 6 7

Sets and σ-Fields

- Sets give probability its structure: outcomes are points, **events are sets**, and probabilities are assigned to sets

σ-FIELD (σ-ALGEBRA)

A collection $\mathcal{F}$ of subsets of Ω is a σ-field if:

$$\text{(i) } \Omega \in \mathcal{F} \qquad \text{(ii) } A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F} \qquad \text{(iii) } A_1, A_2, \ldots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$$

BOREL FIELD

- The **smallest** σ-field on $\mathbb{R}$ that contains all open sets (equivalently, all intervals)
- Closed under **countable unions**, **countable intersections** and **complements**

$$\mathcal{B}(\mathbb{R}) = \sigma\big(\{(a, b] : a < b\}\big)$$

Probability Space

$$(\Omega,\ \mathcal{F},\ P)$$

- Ω Sample space: nonempty set of outcomes (“states of nature”)
- $\mathcal{F}$ Events: subsets of Ω forming a σ-field (Borel field)
- P Probability measure P: $\mathcal{F} \to [0, 1]$

KOLMOGOROV'S AXIOMS AND CONSEQUENCES

$$P(A) \geq 0, \qquad P(\Omega) = 1, \qquad P\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} P(A_i) \ \text{ for disjoint } A_i$$

$$P(\varnothing) = 0, \quad P(A^c) = 1 - P(A), \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Random Variables

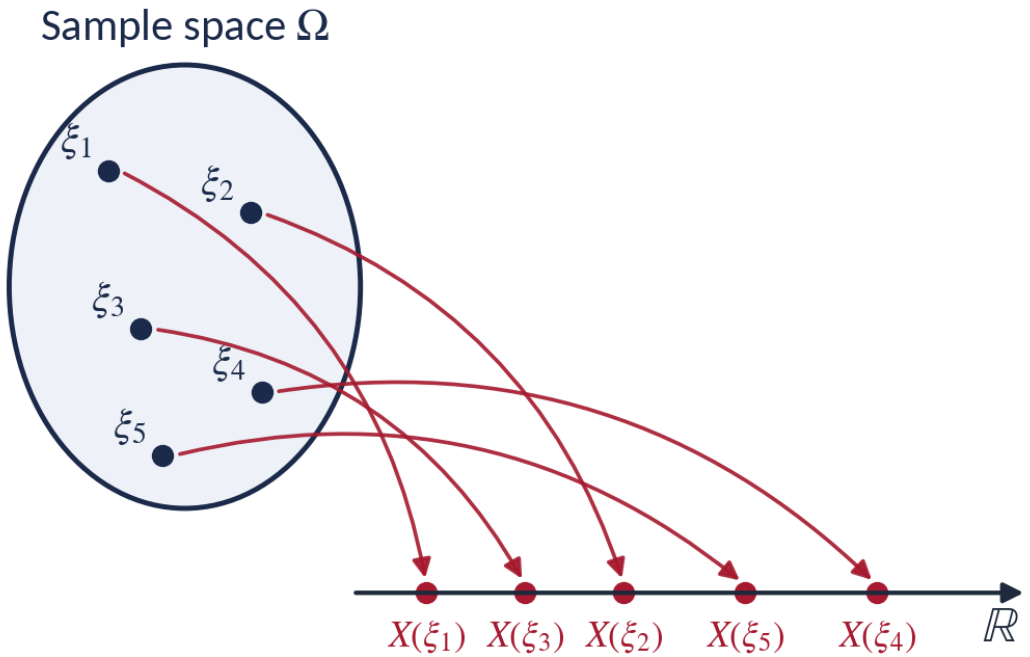
DEFINITION

A **random variable** (RV) is a **function** (mapping) from the outcomes of an experiment to the real (or complex) numbers

$$X : \Omega \to \mathbb{R}, \qquad \xi \mapsto X(\xi)$$

Requirement (measurability), so that P(X ≤ x) is always defined:

$$\{\xi \in \Omega : \ X(\xi) \leq x\} \in \mathcal{F} \quad \text{for every } x \in \mathbb{R}$$



Examples of Random Variables



EXAMPLE I

Faces of a die

Map each face to the first six natural numbers {1, …, 6}

Discrete



EXAMPLE II

Height of a person

Map a person's height to the real interval (0, 3] meters

Continuous



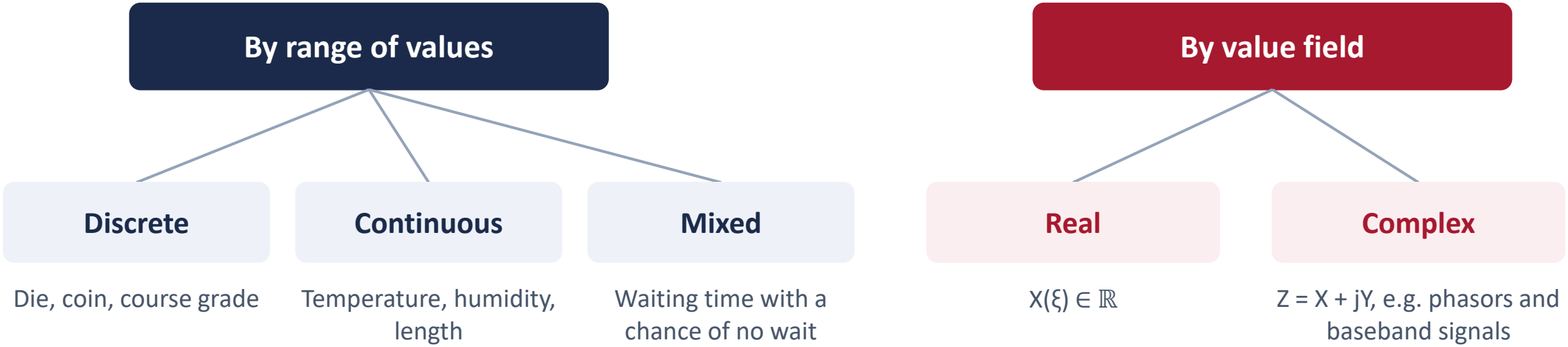
EXAMPLE III

Exam grade

Map exam performance to {0, 0.1, 0.2, …, 20}: steps of 0.1

Discrete

Types of Random Variables



NOTE

A complex RV is a pair of real RVs (X, Y); everything we say about real RVs extends to it through their joint distribution.

03

Density & Distribution Functions

PMF, CDF and PDF; the families you will use all term; expectation.

Probability Mass Function (PMF)

- Used for **discrete** random variables

$$p_X(x_i) = P(X = x_i), \qquad \sum_i p_X(x_i) = 1$$

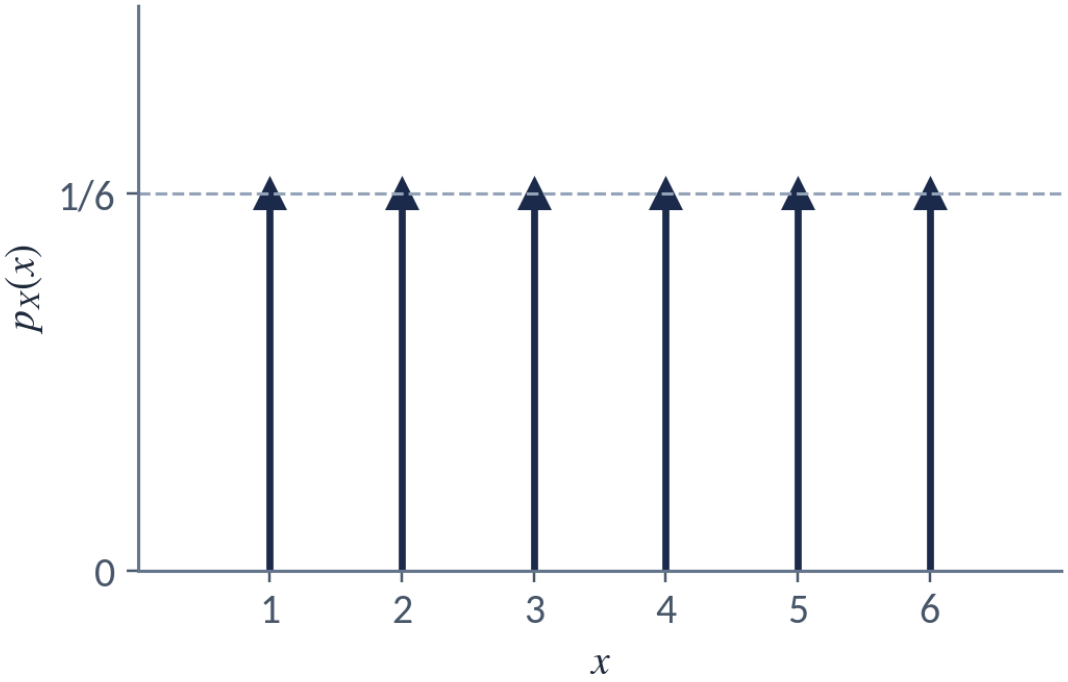
- As a density: a **sum of impulses**; each impulse's magnitude is the probability of that outcome

$$f_X(x) = \sum_i p_X(x_i)\,\delta(x - x_i)$$

EXAMPLE: ROLLING A FAIR DIE

$$f_X(x) = \frac{1}{6}\sum_{i=1}^{6} \delta(x - i)$$

Fair die: six impulses of weight 1/6

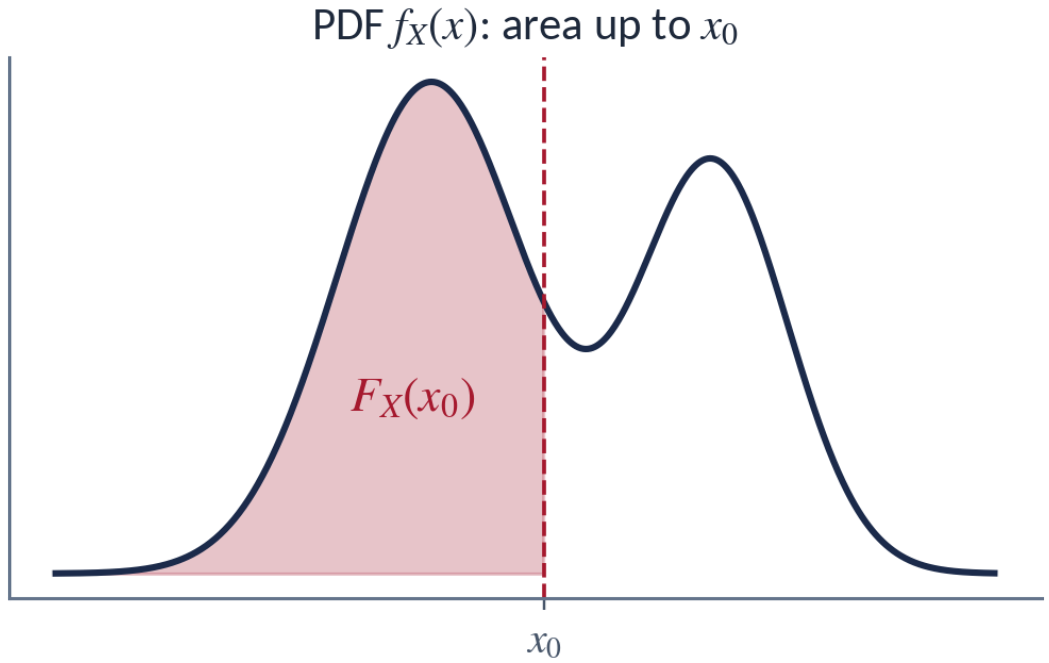


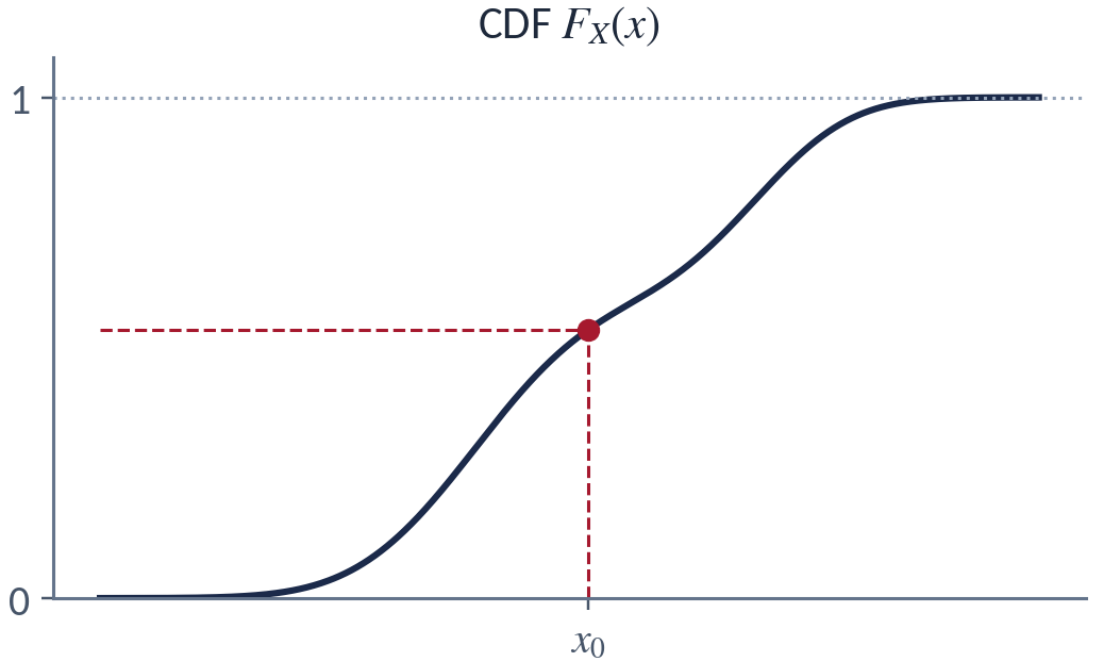
Cumulative Distribution Function (CDF)

DEFINITION

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(u)\, du$$

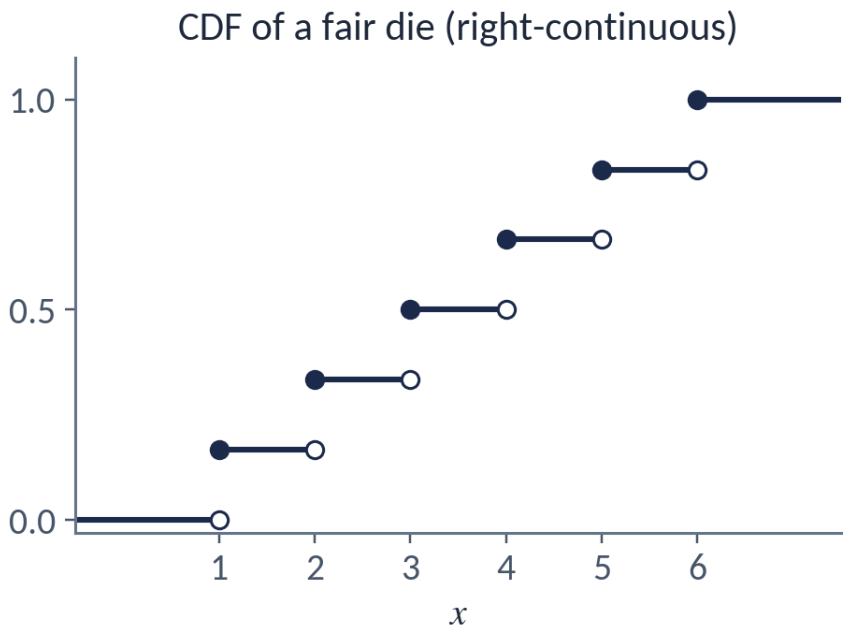
- Defined for **both** continuous and discrete RVs
- The **integral of the PDF**: the area to the left of x





Properties of the CDF

1. **Non-decreasing** $x_1 < x_2 \Rightarrow F_X(x_1) \leq F_X(x_2)$
2. **Right-continuous** $F_X(x^+) = F_X(x)$
3. **Limits** $F_X(-\infty) = 0, \qquad F_X(+\infty) = 1$
4. **Interval probability** $P(a < X \leq b) = F_X(b) - F_X(a)$
5. **Jumps = point masses** $P(X = x) = F_X(x) - F_X(x^-)$



Probability Density Function (PDF)

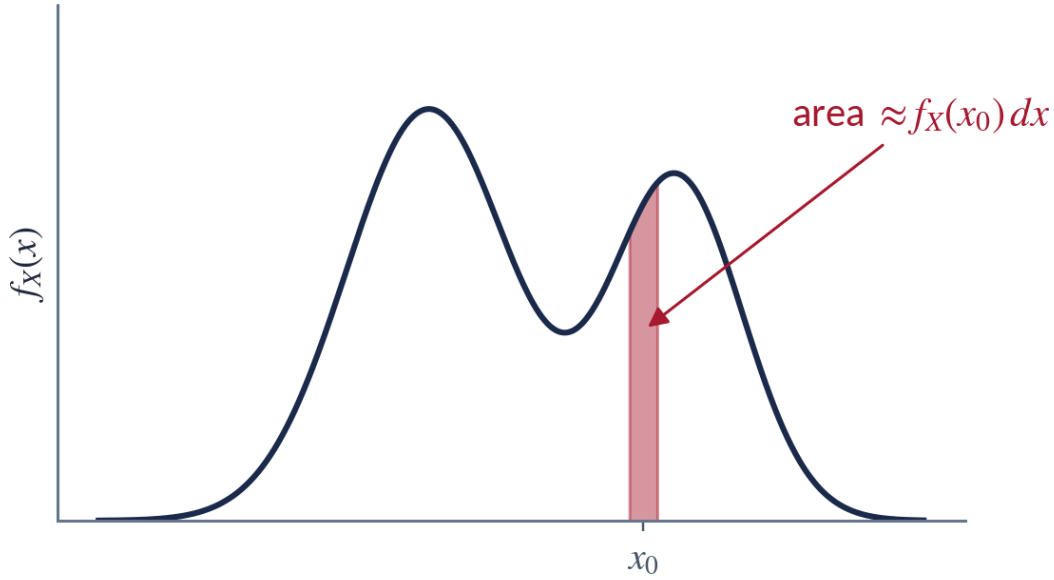
- Used for **continuous** random variables

$$f_X(x) = \frac{dF_X(x)}{dx}$$

$$f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

- Probability of X falling in a small interval around x_0:

$$P\left(x_0 - \tfrac{dx}{2} < X \leq x_0 + \tfrac{dx}{2}\right) \approx f_X(x_0)\,dx$$

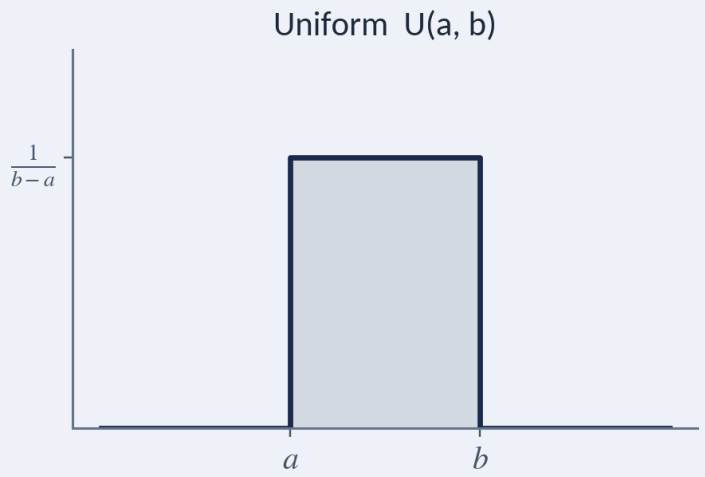


NOTE

$f_X(x)$ is a density, **not** a probability: it can exceed 1.

Common Densities: Uniform and Gaussian

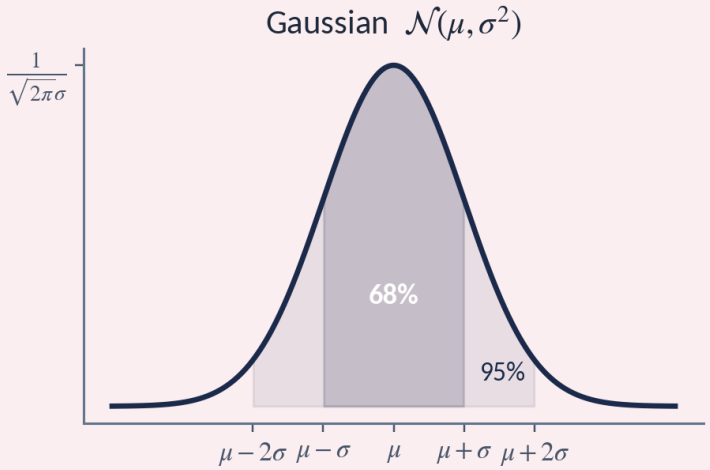
UNIFORM U(a, b)



$$f_X(x) = \frac{1}{b-a}\big[U(x-a) - U(x-b)\big]$$

$$E[X] = \frac{a+b}{2}, \qquad \mathrm{Var}[X] = \frac{(b-a)^2}{12}$$

GAUSSIAN (NORMAL)

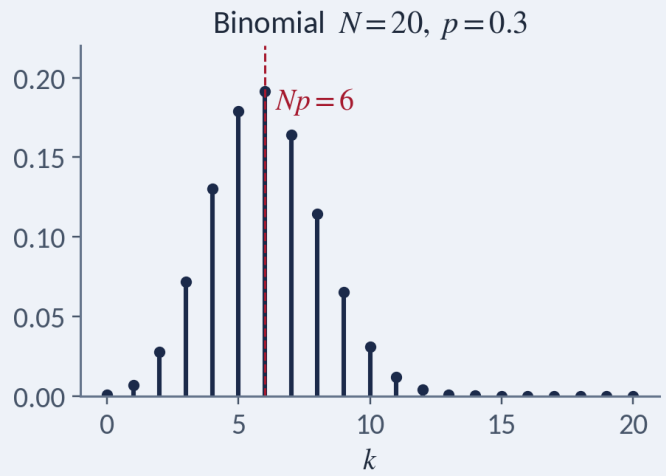


$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}} \equiv \mathcal{N}(\mu, \sigma^2)$$

$$E[X] = \mu, \qquad \mathrm{Var}[X] = \sigma^2$$

Discrete Families: Binomial and Poisson

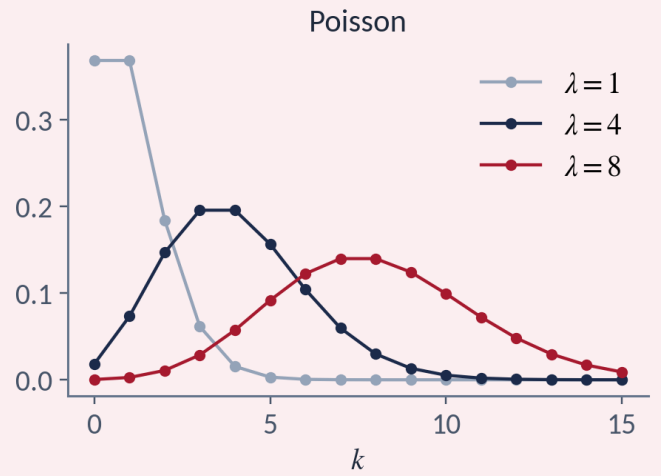
BINOMIAL



$$P(X=k)=\binom{N}{k}p^k(1-p)^{N-k},\quad k=0,\dots,N$$

$$E[X]=Np,\qquad \mathrm{Var}[X]=Np(1-p)$$

POISSON



$$P(X=k)=e^{-\lambda}\frac{\lambda^k}{k!},\quad k=0,1,2,\dots$$

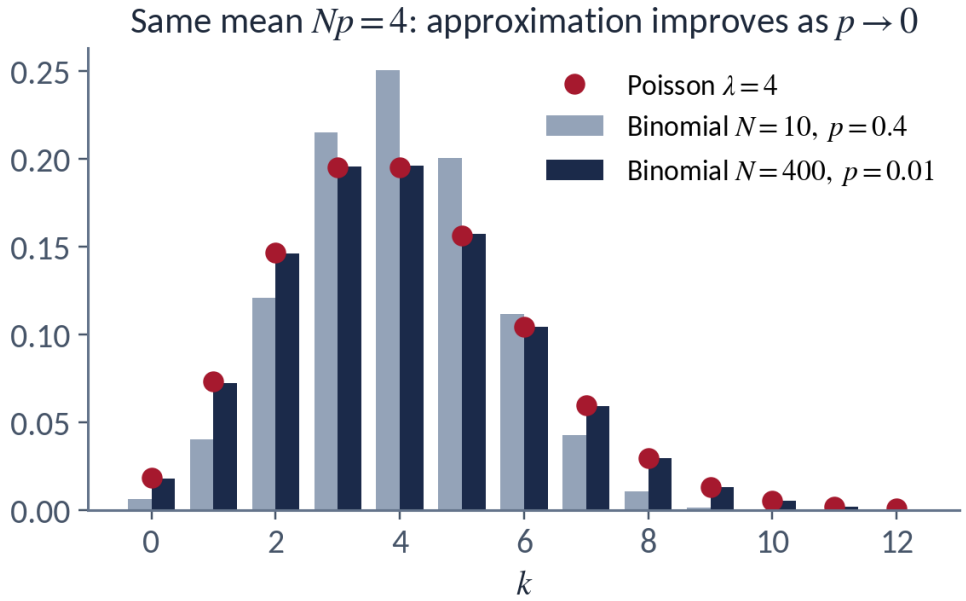
$$E[X]=\mathrm{Var}[X]=\lambda$$

Poisson Approximation to the Binomial

IMPORTANT FACT

$$\binom{N}{k} p^k (1-p)^{N-k} \xrightarrow[N\to\infty,\ p\to 0,\ Np=\lambda]{} e^{-\lambda}\frac{\lambda^k}{k!}$$

- **Rare events in many trials:** replace Binomial(N, p) by Poisson(λ = Np)
- Needs **N large and p small**; large N alone is not enough (compare the grey bars)



Exponential Density

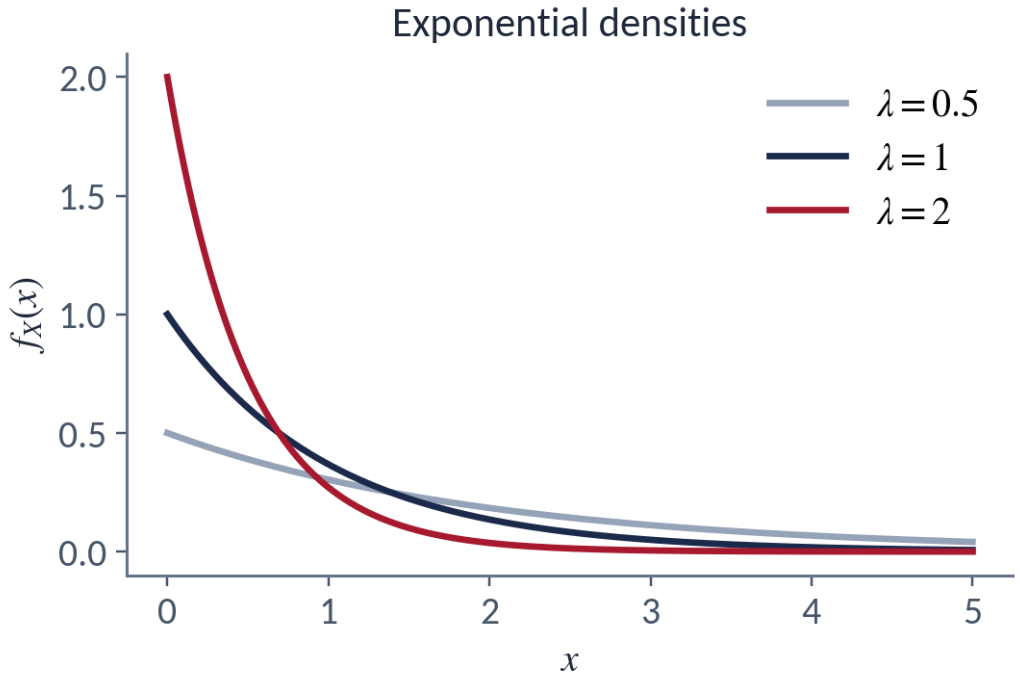
$$f_X(x) = \lambda e^{-\lambda x}\, U(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$F_X(x) = 1 - e^{-\lambda x}\ (x \geq 0), \qquad E[X] = \frac{1}{\lambda}, \qquad \mathrm{Var}[X] = \frac{1}{\lambda^2}$$

MEMORYLESS PROPERTY

$$P(X > s + t \mid X > s) = P(X > t)$$

The only continuous memoryless distribution; it models waiting times between Poisson arrivals



Expected Value

MEAN (EXPECTED VALUE)

$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx \qquad E[X] = \sum_i x_i\, p_X(x_i)$$

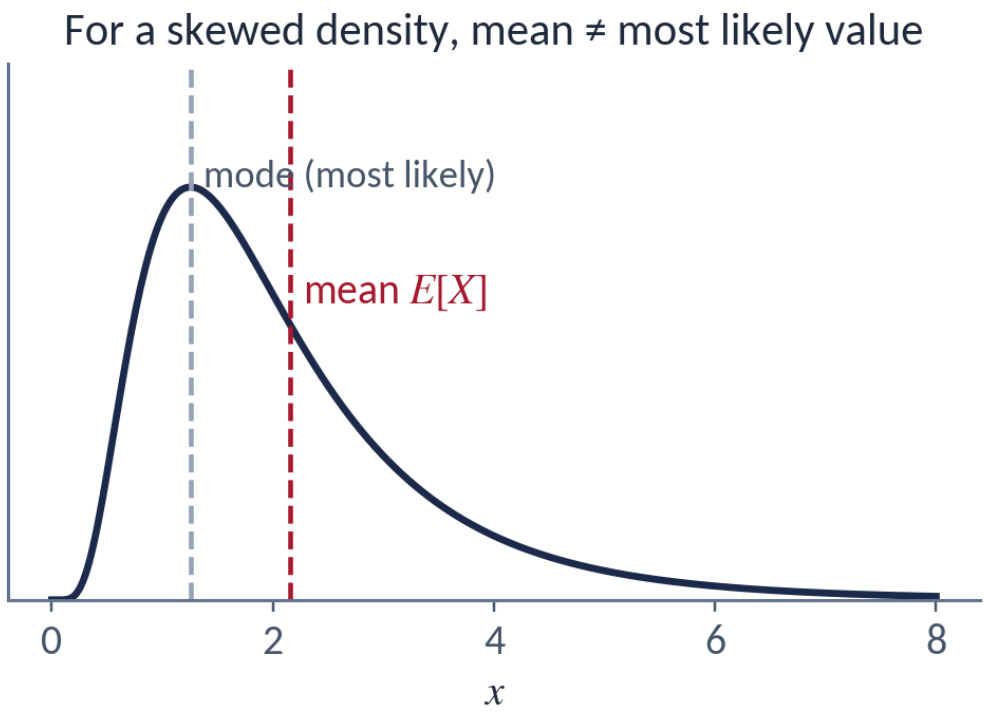
The **center of mass** of the density: not necessarily the most likely value (the mode)

LINEAR OPERATOR

$$E[aX + b] = a\, E[X] + b$$

FUNCTION OF AN RV

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx$$



PDF of a Function of a Random Variable

THEOREM

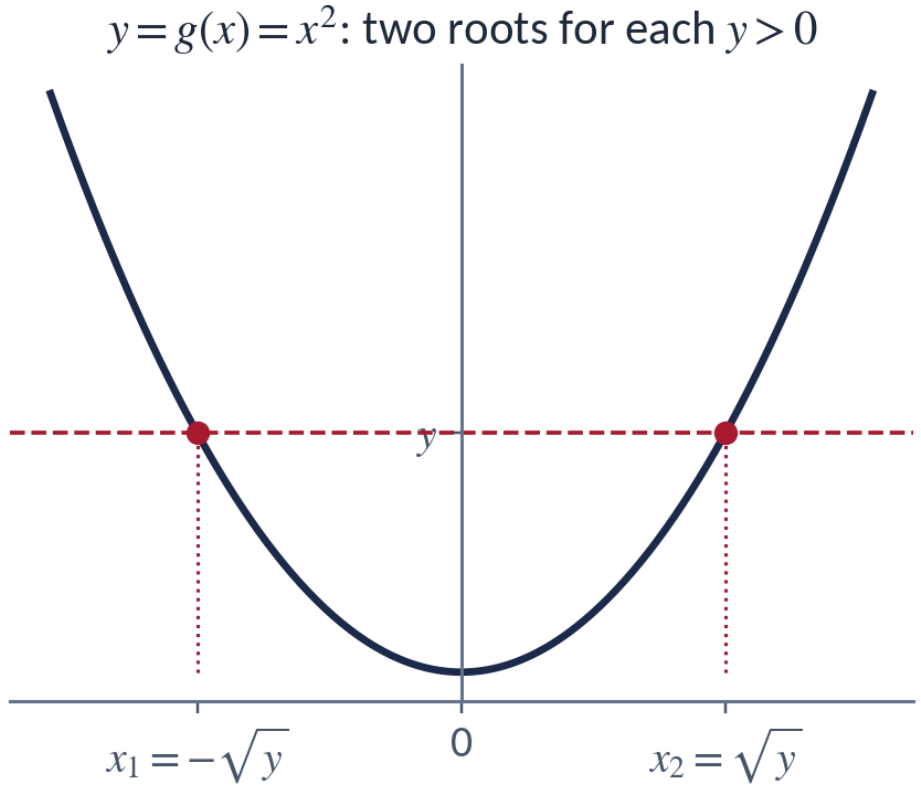
Let Y = g(X), and suppose y = g(x) has the real roots x_1, …, x_n:

$$y = g(x) \;\Rightarrow\; x_1, x_2, \ldots, x_n$$

Then the PDF of Y follows from the PDF of X:

$$f_Y(y) = \sum_{i=1}^{n} \frac{f_X(x_i)}{|g'(x_i)|}$$

Sum over **all** real roots; if y = g(x) has no root, $f_Y(y) = 0$



Worked Examples

LINEAR TRANSFORMATION

$$Y = aX + b \;\Rightarrow\; f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

SQUARE LAW (TWO ROOTS)

$$Y = X^2 \;\Rightarrow\; f_Y(y) = \frac{f_X(\sqrt{y}) + f_X(-\sqrt{y})}{2\sqrt{y}}\, U(y)$$

GAUSSIAN SQUARED: CHI-SQUARE, 1 DEGREE OF FREEDOM

$$X \sim \mathcal{N}(0,1) \;\Rightarrow\; f_Y(y) = \frac{1}{\sqrt{2\pi y}}\, e^{-y/2}\, U(y) \quad (\chi^2_1)$$

04

Joint & Conditional Distributions

Two or more random variables at once: joint, marginal and conditional descriptions.

1 2 3 4 5 6 7

Joint Distribution and Density

JOINT DISTRIBUTION FUNCTION

$$F_{X,Y}(x,y) = P(X \le x,\ Y \le y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,dv\,du$$

JOINT DENSITY FUNCTION

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x\,\partial y}$$

PROBABILITY OF A REGION

$$P\big((X,Y) \in D\big) = \iint_{D} f_{X,Y}(x,y)\,dx\,dy$$

- $f_{X,Y}(x, y) \ge 0$ and it integrates to 1 over the whole plane
- Everything about the pair (X, Y) is contained in $F_{X,Y}$ (or $f_{X,Y}$)

Example: A Die as a 3-Bit Number

Roll a fair die and write the outcome as a 3-bit number “xyz”. Let X and Y be the first two bits.

Outcome	x y z
1	0 0 1
2	0 1 0
3	0 1 1
4	1 0 0
5	1 0 1
6	1 1 0

Joint PMF $p_{X,Y}(x, y)$

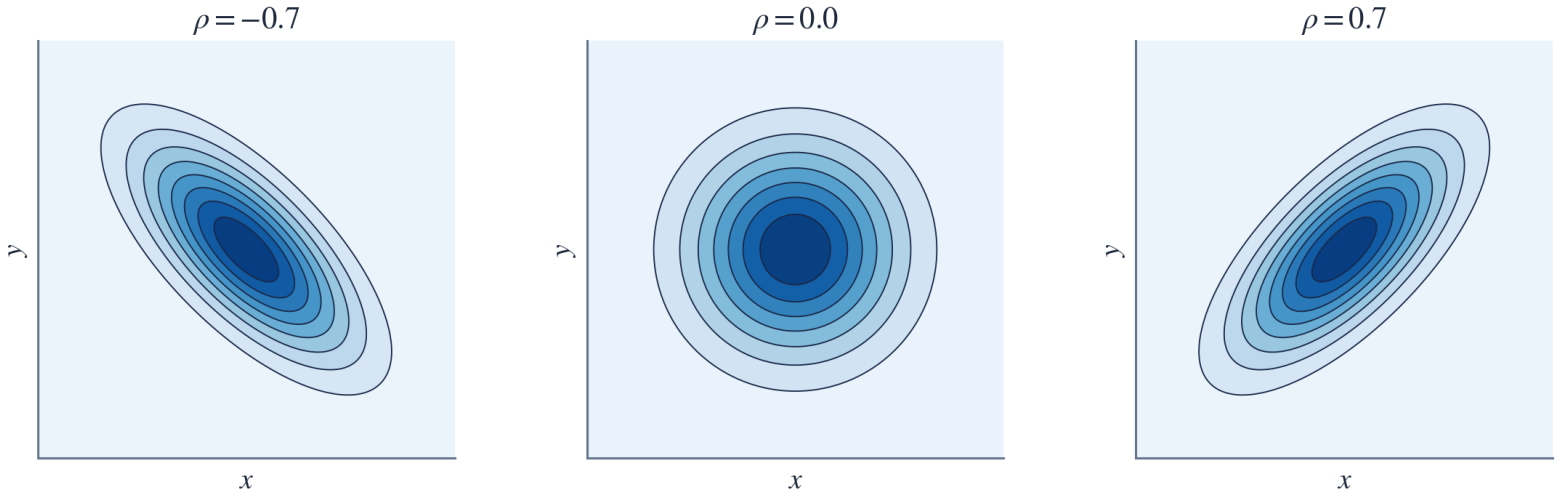
	y = 0	y = 1
x = 0	1/6 {1}	1/3 {2, 3}
x = 1	1/3 {4, 5}	1/6 {6}

ARE X AND Y INDEPENDENT?

P(X = 0) = P(Y = 0) = 1/2, but P(X = 0, Y = 0) = 1/6 ≠ 1/4. **No:** the joint PMF does not factor.

Jointly Gaussian Random Variables

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]\right\}$$



WHAT IS ρ?

- ρ is the **correlation coefficient** of X and Y (Section 5): the strength and direction of their linear relationship
- If ρ = 0, the density factors into $f_X(x)\, f_Y(y)$

Marginal Densities

- Get the density of **one** variable by integrating out the other:

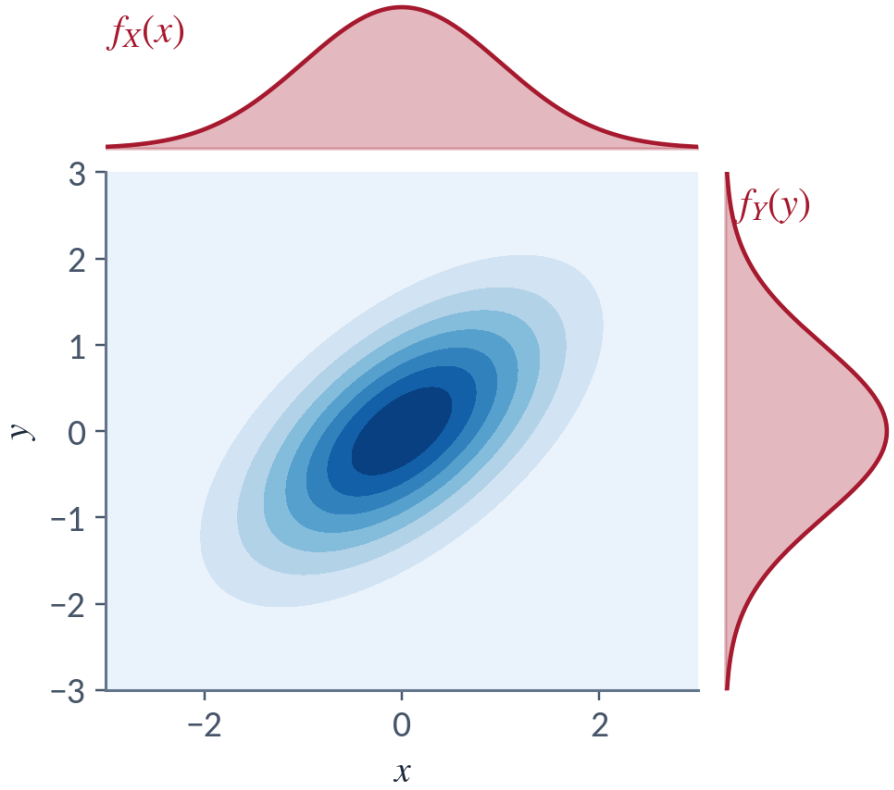
$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx$$

HOW DO WE GET THE DISTRIBUTION FUNCTIONS?

Integrate the marginal density, or let the other variable go to infinity:

$$F_X(x) = F_{X,Y}(x, +\infty) = \int_{-\infty}^{x} f_X(u)\, du$$



Conditional Density and Bayes' Rule

CONDITIONAL DENSITY

Density of X once we have **observed** Y = y:

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}, \qquad f_Y(y) > 0$$

BAYES' RULE

$$f_{X|Y}(x \mid y) = \frac{f_{Y|X}(y \mid x)\, f_X(x)}{\int_{-\infty}^{\infty} f_{Y|X}(y \mid x)\, f_X(x)\, dx}$$

Prior $f_X(x)$ × likelihood $f_{Y|X}(y|x)$ → posterior $f_{X|Y}(x|y)$

Examples of Conditioning

EXAMPLE I: ROLLING A FAIR DIE

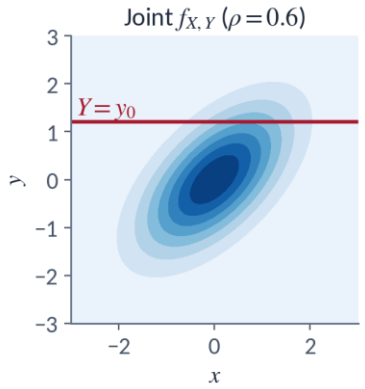
- X: the outcome is **even** {2, 4, 6}
- Y: the outcome is **prime** {2, 3, 5}

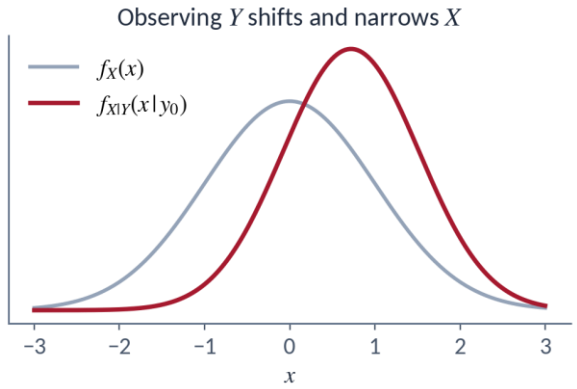
$$P(X \mid Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{P(\{2\})}{P(\{2,3,5\})} = \frac{1/6}{1/2} = \frac{1}{3}$$

EXAMPLE II: JOINTLY GAUSSIAN X, Y

Given Y = y, X is still Gaussian, with a shifted mean and a smaller variance:

$$X \mid Y = y \;\sim\; \mathcal{N}\!\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y-\mu_Y),\; \sigma_X^2(1-\rho^2)\right)$$





Conditional Distribution Function

DEFINITION

$$F_{X|Y}(x \mid y) = P(X \leq x \mid Y = y) = \int_{-\infty}^{x} f_{X|Y}(u \mid y)\, du = \frac{\int_{-\infty}^{x} f_{X,Y}(u, y)\, du}{\int_{-\infty}^{\infty} f_{X,Y}(u, y)\, du}$$

- Note that **y is a constant** during the integration
- The denominator is just $f_Y(y)$: it normalizes the slice of the joint density at Y = y (see the previous figure)

Independent Random Variables

DEFINITION

$$F_{X,Y}(x,y) = F_X(x)F_Y(y) \iff f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

CONSEQUENCE: OBSERVING Y TELLS US NOTHING ABOUT X

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{f_X(x)\, f_Y(y)}{f_Y(y)} = f_X(x)$$

FUNCTIONS OF INDEPENDENT RVs

$$X, Y \text{ indep.} \Rightarrow g(X),\ h(Y) \text{ indep.}, \quad E[g(X)h(Y)] = E[g(X)]\, E[h(Y)]$$

Functions of Two Random Variables

Assume $U = g(X, Y), \qquad V = h(X, Y)$

The inverse system has solutions $(u, v) \Rightarrow (x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$

Define the **Jacobian** matrix

$$J(x, y) = \begin{bmatrix} \dfrac{\partial g}{\partial x} & \dfrac{\partial g}{\partial y} \\ \dfrac{\partial h}{\partial x} & \dfrac{\partial h}{\partial y} \end{bmatrix}$$

THE JOINT PDF

$$f_{U,V}(u, v) = \sum_{i=1}^{n} \frac{f_{X,Y}(x_i, y_i)}{|\det J(x_i, y_i)|}$$

Example: Cartesian to Polar Coordinates

$$X, Y \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \qquad R = \sqrt{X^2 + Y^2}, \quad \Theta = \operatorname{atan2}(Y, X)$$

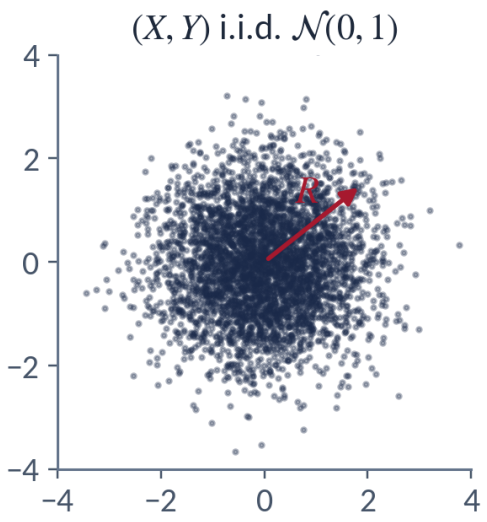
$$x = r\cos\theta,\ y = r\sin\theta \ \Rightarrow\ |\det J| = \frac{1}{r}$$

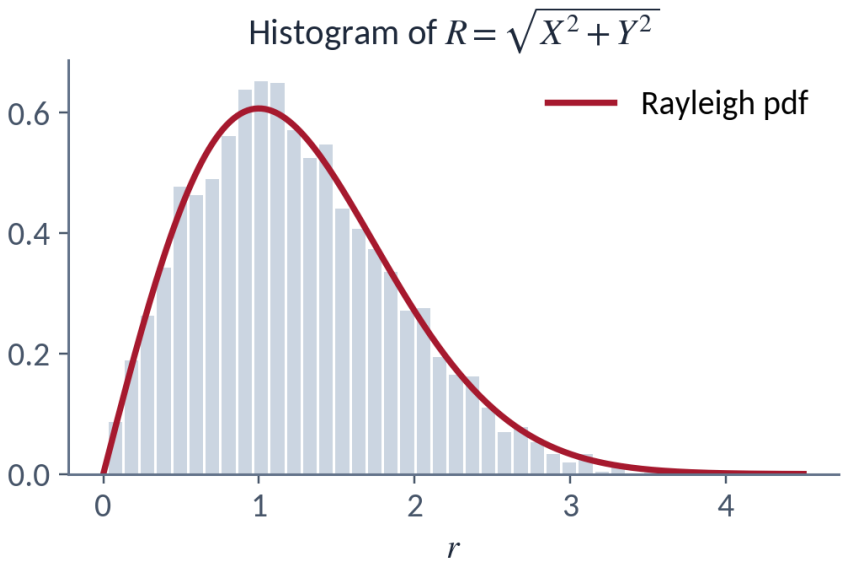
$$f_{R,\Theta}(r,\theta) = r\, f_{X,Y}(r\cos\theta, r\sin\theta) = \frac{r}{2\pi\sigma^2}\, e^{-r^2/2\sigma^2}$$

RESULT: R AND Θ ARE INDEPENDENT

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/2\sigma^2} U(r) \quad \text{(Rayleigh)}$$

$$f_\Theta(\theta) = \frac{1}{2\pi}, \quad 0 < \theta \le 2\pi \quad \text{(Uniform)}$$





05

Linear Correlation

How much does knowing X tell us about Y, linearly?

1 2 3 4 5 6 7

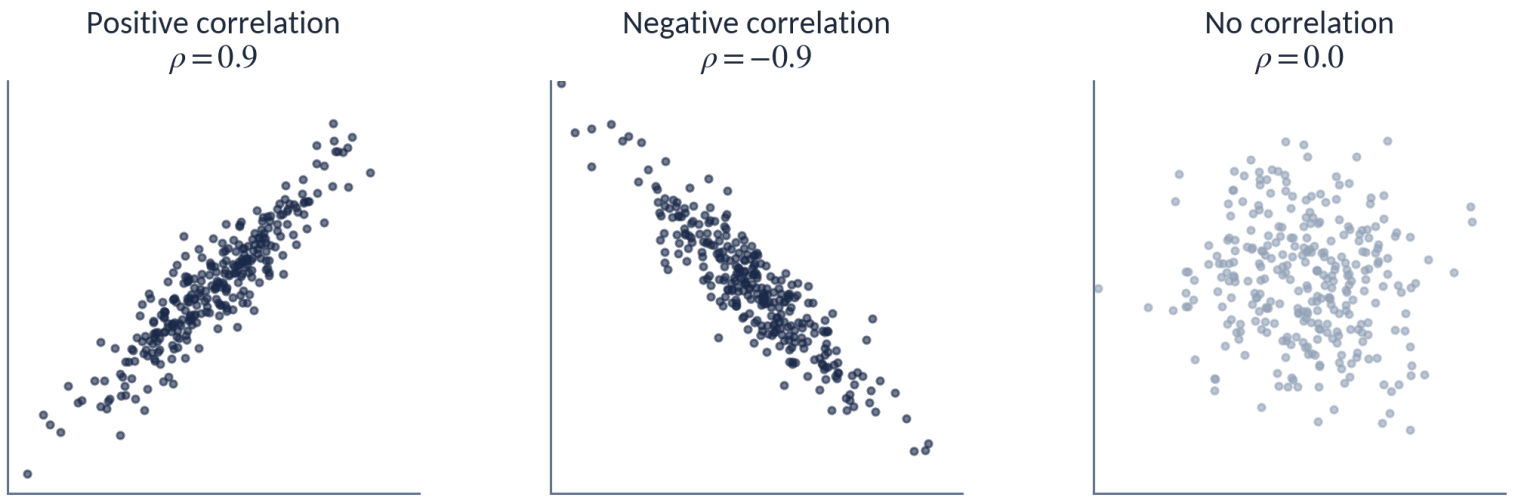
Covariance and Correlation Coefficient

Knowing X, how much do we learn about Y? Covariance measures how two variables **change together** (linear similarity).

$$\mathrm{Cov}(X,Y) = E\big[(X-\mu_X)(Y-\mu_Y)\big] = E[XY] - \mu_X\mu_Y$$

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\,\sigma_Y}, \qquad -1 \le \rho_{XY} \le 1$$

ρ is dimensionless; |ρ| = 1 exactly when Y = aX + b.



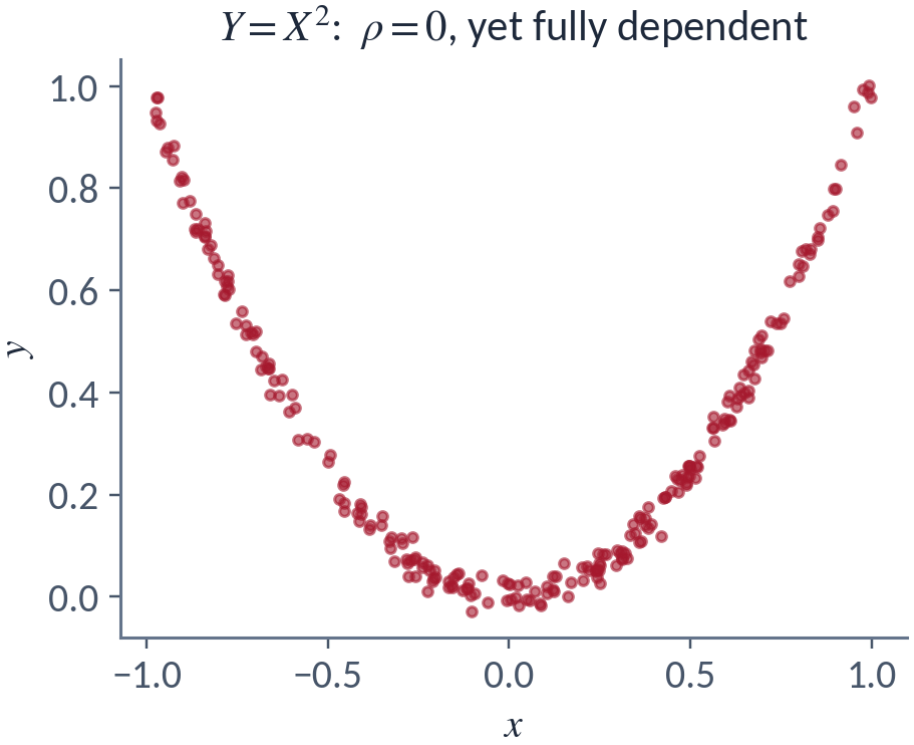
Uncorrelated Is Not the Same as Independent

$$\mathrm{Cov}(X,Y) = 0 \iff E[XY] = E[X]\,E[Y]$$

- Zero covariance does **not** imply independence
- Independence is stronger: **no relationship of any kind**, not just linear
- Independent ⇒ uncorrelated (the converse fails, except for jointly Gaussian RVs)

COUNTEREXAMPLE

$$X \sim U(-1,1),\ Y = X^2: \quad \mathrm{Cov}(X,Y) = E[X^3] - E[X]E[X^2] = 0$$



Variance, Correlation and Standard Deviation

VARIANCE: COVARIANCE OF X WITH ITSELF

$$\mathrm{Var}(X) = \sigma_X^2 = \mathrm{Cov}(X, X) = E\big[(X - \mu_X)^2\big]$$

$$E[X^2] = \sigma_X^2 + \mu_X^2 \quad \Longleftrightarrow \quad \sigma_X^2 = E[X^2] - \mu_X^2$$

CORRELATION VS. COVARIANCE

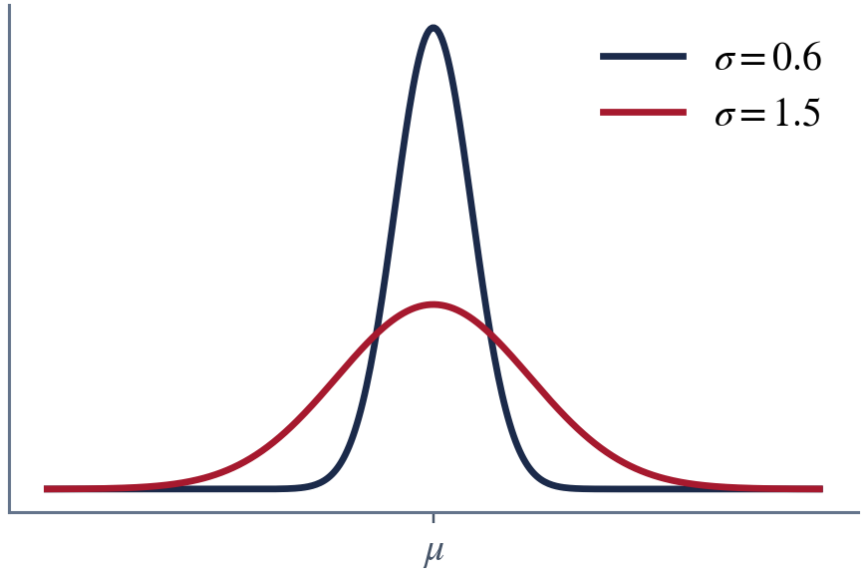
$$R_{XY} = E[XY], \qquad \mathrm{Cov}(X, Y) = R_{XY} - \mu_X \mu_Y$$

$$R_{XY} = 0 \ \ (\text{orthogonal}), \qquad \mathrm{Cov}(X, Y) = 0 \ \ (\text{uncorrelated})$$

STANDARD DEVIATION

$\sigma_X = \sqrt{\mathrm{Var}(X)}$: same units as X

Same mean, different spread



Moments

n-TH MOMENT

$$m_n = E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x)\, dx$$

n-TH CENTRAL MOMENT

$$\mu_n = E\big[(X - \mu_X)^n\big]$$

MEAN AND VARIANCE

$$m_1 = \mu_X, \qquad \mu_2 = \sigma_X^2 = m_2 - m_1^2$$

The mean is the first moment; the variance is the **second moment minus the squared mean**

NORMALIZED (STANDARDIZED) MOMENTS

$$\gamma_1 = \frac{\mu_3}{\sigma_X^3}\ (\text{skewness}), \qquad \gamma_2 = \frac{\mu_4}{\sigma_X^4}\ (\text{kurtosis}; = 3 \text{ for Gaussian})$$

06

Important Theorems

Limit theorems and inequalities that make averages trustworthy.

Central Limit Theorem (CLT)

THEOREM

$X_1, X_2, \ldots$ i.i.d. with mean μ and **finite** variance σ^2:

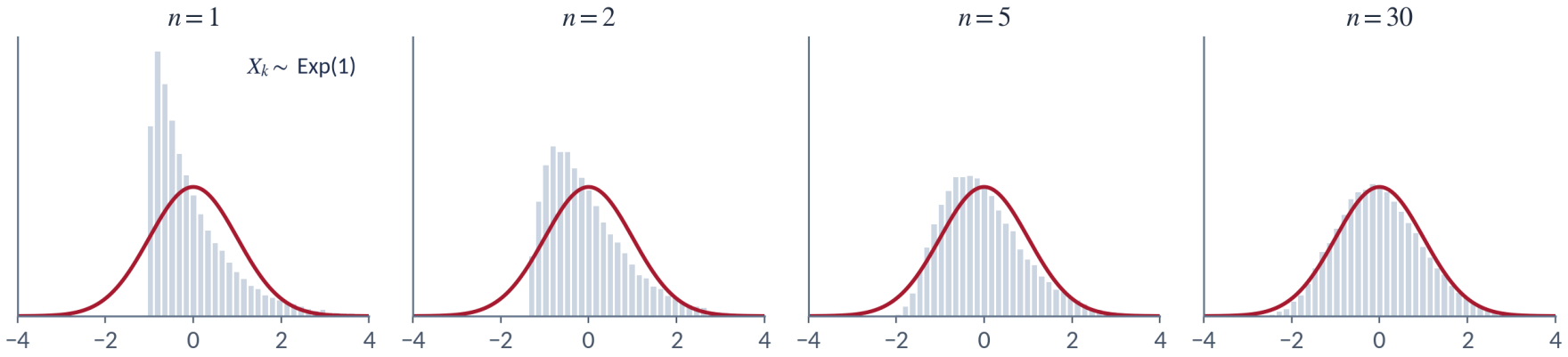
$$S_n = \sum_{k=1}^{n} X_k, \qquad Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

$$\lim_{n\to\infty} P(Z_n \le z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du$$

EXCEPTION: CAUCHY. WHY?

Cauchy RVs have no mean and infinite variance, so the hypothesis fails; the average of n Cauchy RVs is again Cauchy

The normalized sum approaches N(0, 1) whatever the distribution of X_k:



Weak Law of Large Numbers

- For i.i.d. RVs X_k with mean μ_X, the **sample mean**

$$\bar{X}_n = \frac{1}{n}\sum_{k=1}^{n} X_k$$

CONVERGES IN PROBABILITY

$$\forall \varepsilon > 0: \qquad \lim_{n\to\infty} P\big(|\bar{X}_n - \mu_X| > \varepsilon\big) = 0$$

PROOF (WHEN σ^2 IS FINITE), VIA CHEBYSHEV

$$P\big(|\bar{X}_n - \mu_X| > \varepsilon\big) \le \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma_X^2}{n\varepsilon^2} \xrightarrow[n\to\infty]{} 0$$

Strong Law of Large Numbers

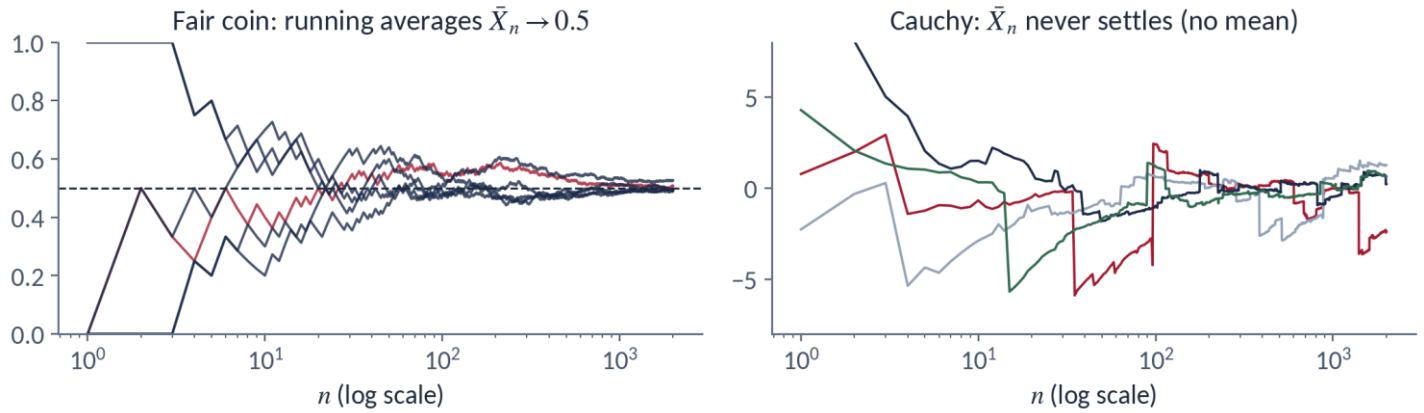
CONVERGES ALMOST SURELY

$$P\left(\lim_{n\to\infty} \bar{X}_n = \mu_X\right) = 1$$

WHY IS THIS STRONGER THAN THE WEAK LAW?

Almost-sure convergence implies convergence in probability, not the other way round

- **Weak:** for each large n, a big deviation is unlikely, but individual sequences may still stray again and again
- **Strong:** almost every sequence of running averages eventually stays close to μ_X for good



Markov and Chebyshev Inequalities

MARKOV'S INEQUALITY ("CHEBYSHEV'S FIRST INEQUALITY")

$$P(X \geq c) \leq \frac{E[X]}{c}, \qquad X \geq 0,\ c > 0$$

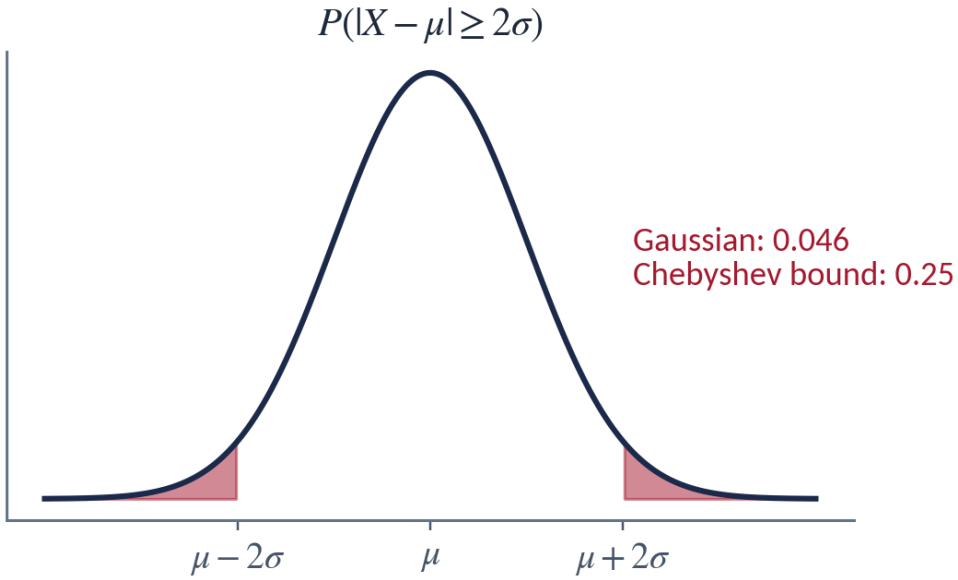
$$E[X] \geq \int_c^\infty x f_X(x)\,dx \geq c \int_c^\infty f_X(x)\,dx = c\,P(X \geq c)$$

CHEBYSHEV'S INEQUALITY

$$P\big(|X - \mu_X| \geq \varepsilon\big) \leq \frac{\sigma_X^2}{\varepsilon^2}$$

Apply Markov to $(X - \mu_X)^2$ with $c = \varepsilon^2$. With $\varepsilon = k\sigma$:

$$\varepsilon = k\sigma_X: \quad P\big(|X - \mu_X| \geq k\sigma_X\big) \leq \frac{1}{k^2}$$



Cauchy–Schwarz Inequality

FOR RVs X, Y WITH FINITE SECOND MOMENTS

$$\big(E[XY]\big)^2 \le E[X^2]\,E[Y^2]$$

Equality holds **iff** X = aY with probability 1 (linear dependence)

PROOF

$$0 \le E\big[(X - aY)^2\big] = E[X^2] - 2a\,E[XY] + a^2 E[Y^2] \quad \forall a \in \mathbb{R}$$

A quadratic in a that is never negative has a non-positive discriminant.

CONSEQUENCE

$$\big(\mathrm{Cov}(X,Y)\big)^2 \le \sigma_X^2\,\sigma_Y^2 \;\Rightarrow\; |\rho_{XY}| \le 1$$

07

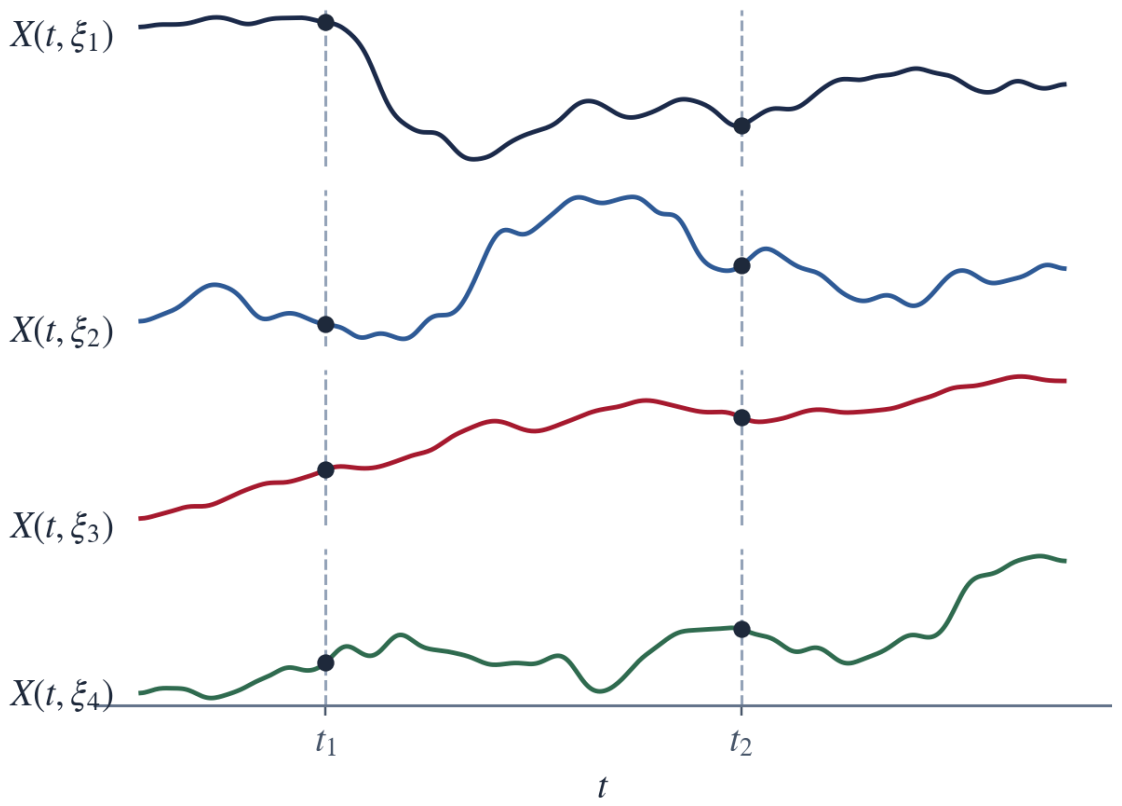
Introduction to Stochastic Processes

A random variable for every instant of time: random functions.

1 2 3 4 5 6 7

What Is a Stochastic Process?

- Let ξ denote the **random outcome** of an experiment
- To every outcome, assign a **function of time** X(t, ξ)
- The collection of such functions is a **stochastic process**
- Both ξ and the time index t can be **continuous or discrete** (countably infinite or finite)
- For a fixed $\xi_i \in \Omega$ (the set of all outcomes), $X(t, \xi_i)$ is a **specific time function**



Four Ways to Read X(t, ξ)

T FIXED, Ξ FIXED

$X(t_1, \xi_1)$

A **number**

T VARIES, Ξ FIXED

$X(t, \xi_1)$

A **time function** (sample path, realization)

T FIXED, Ξ VARIES

$X_1 = X(t_1, \xi)$

A **random variable**

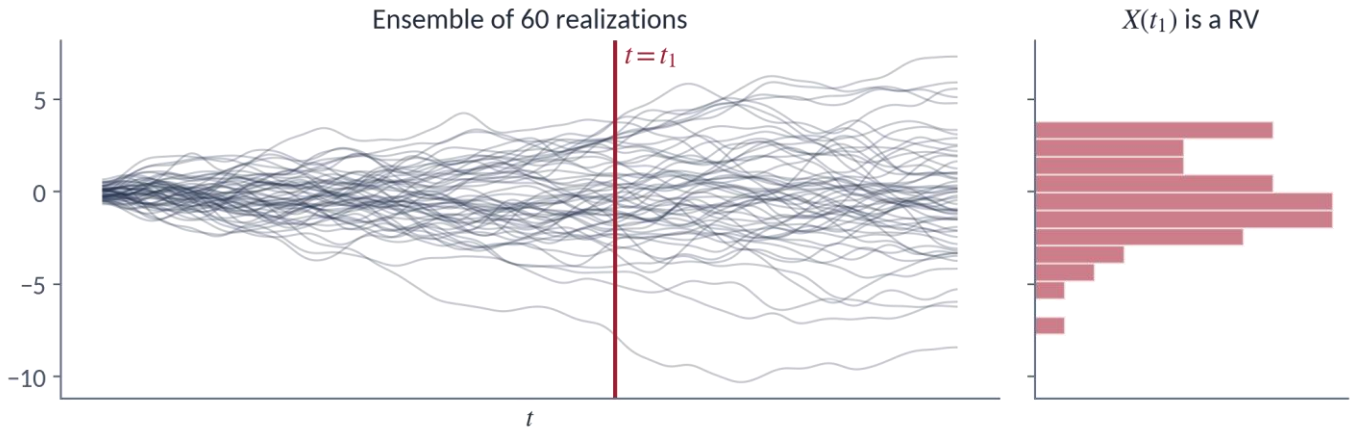
T VARIES, Ξ VARIES

$X(t) = X(t, \xi)$

The **stochastic process** (the ensemble)

Fixed t: A Random Variable

For fixed t, $X_1 = X(t_1, \xi)$ is a **random variable**; the **ensemble** of all realizations over time is the process X(t)



FIRST-ORDER DESCRIPTION

$$F_X(x,t) = P\big(X(t) \le x\big), \qquad f_X(x,t) = \frac{\partial F_X(x,t)}{\partial x}$$

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x\, f_X(x,t)\, dx$$

FULL DESCRIPTION

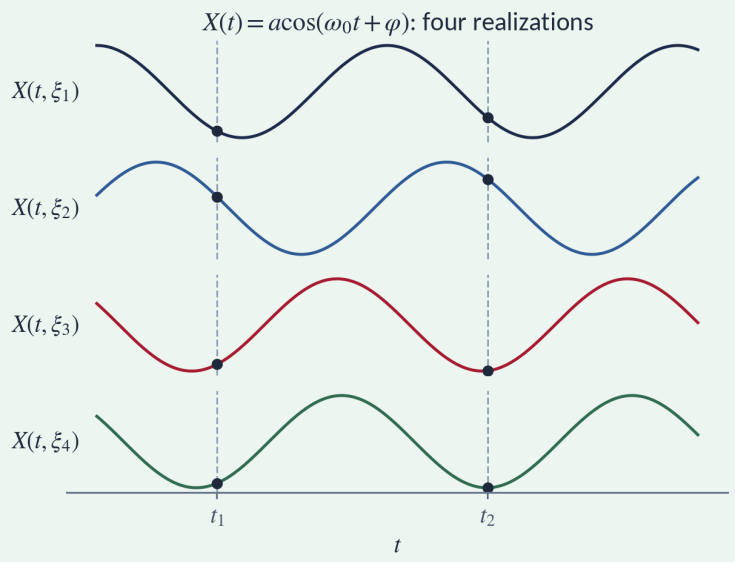
All n-th order distributions, for every n and every choice of $t_1, \ldots, t_n$

Stochastic Processes Are Everywhere

A FIRST EXAMPLE

$$X(t) = a\cos(\omega_0 t + \varphi), \qquad \varphi \sim U(0, 2\pi)$$

ϕ is uniformly distributed in (0, 2π): every outcome gives a sinusoid with a different phase



Stock market fluctuations

Queuing systems

Earthquake signals

1-D audio

2-D images

3-D video

Example 1: X(t) = At + b, First-Order PDF

PROBLEM

$$X(t) = A\,t + b, \qquad A \sim \mathcal{N}(0,1), \quad b \text{ constant}, \quad t > 0$$

Find $f_X(x, t)$

Use the **PDF of a function of an RV** (Section 3) with t held fixed:

$$f_A(a) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{a^2}{2}\right) \qquad X = At + b \;\Rightarrow\; a = \frac{x-b}{t}, \qquad \left|\frac{dX}{dA}\right| = t$$

RESULT

$$f_X(x,t) = \frac{1}{t}\, f_A\left(\frac{x-b}{t}\right) = \frac{1}{\sqrt{2\pi}\,t} \exp\left(-\frac{(x-b)^2}{2t^2}\right) \;=\; \mathcal{N}(b,\, t^2)$$

Example 1: Mean and Variance

Mean

$$E[X(t)] = E[A]\,t + b = 0 \cdot t + b = b$$

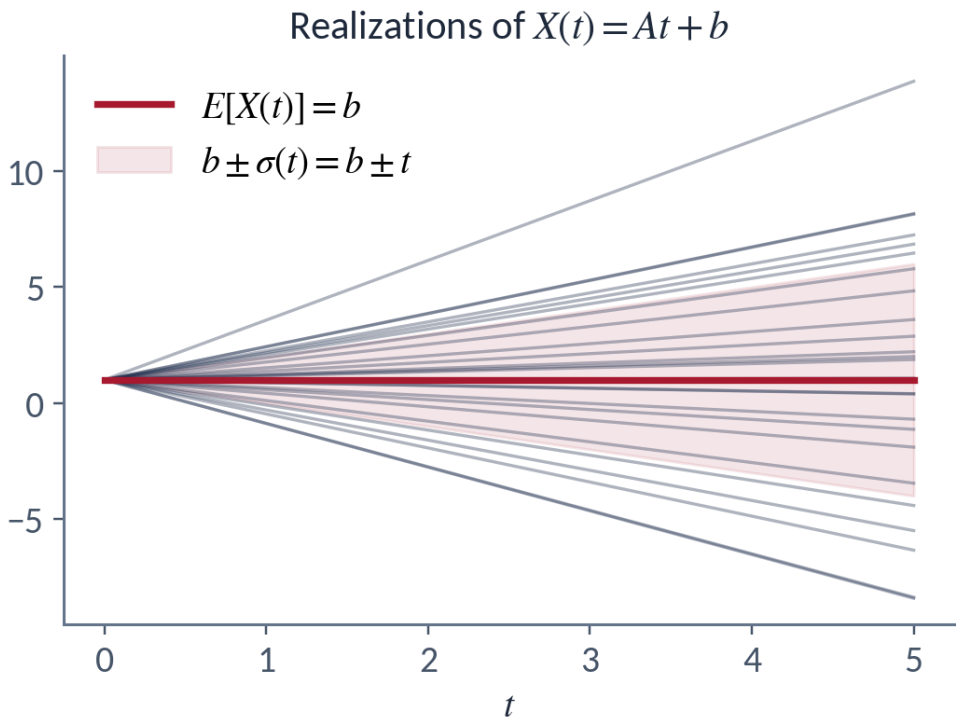
Variance

$$E[X^2(t)] = E[A^2]\,t^2 + 2bt\,E[A] + b^2 = t^2 + b^2$$

$$\mathrm{Var}[X(t)] = E[X^2(t)] - \big(E[X(t)]\big)^2 = t^2$$

NOTE

The **mean is constant**, but the **variance grows with t**: this process is **not stationary**



Example 2: Random-Phase Sinusoid

PROBLEM

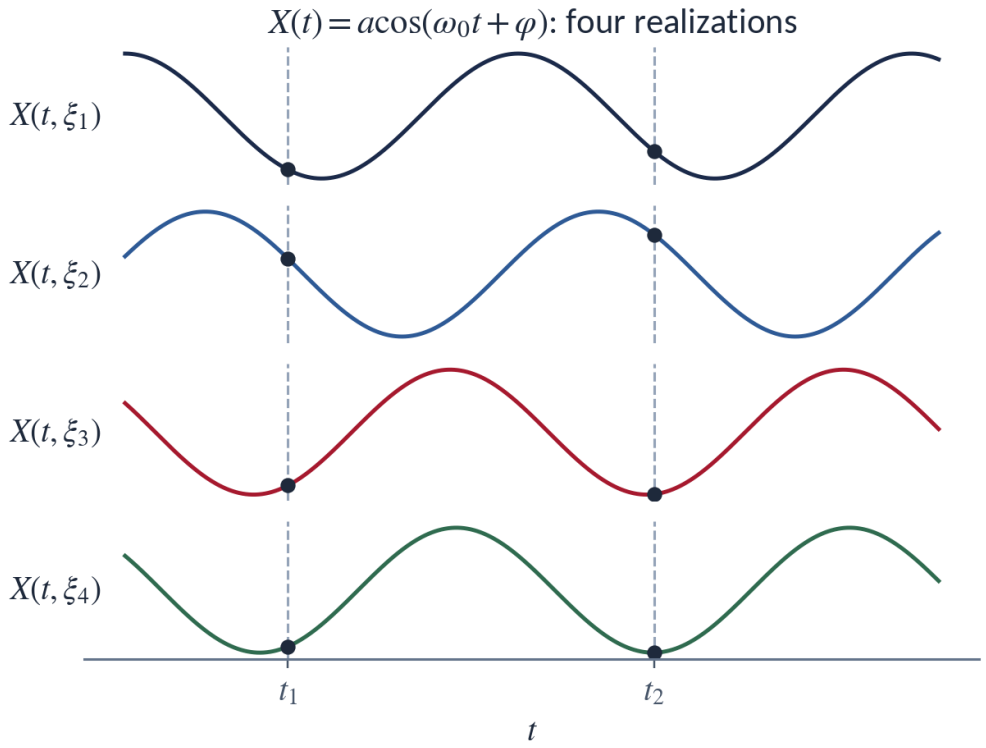
$$X(t) = A\cos(\omega_0 t + \theta), \qquad \theta \sim U(0, 2\pi), \quad A, \omega_0 \text{ constants}$$

$$f_\theta(\theta) = \begin{cases} \frac{1}{2\pi} & \theta \in (0, 2\pi] \\ 0 & \text{else} \end{cases}$$

a) First-order PDF $f_X(x, t)$ = ?

b) E[X(t)] = ?

c) Var[X(t)] = ?



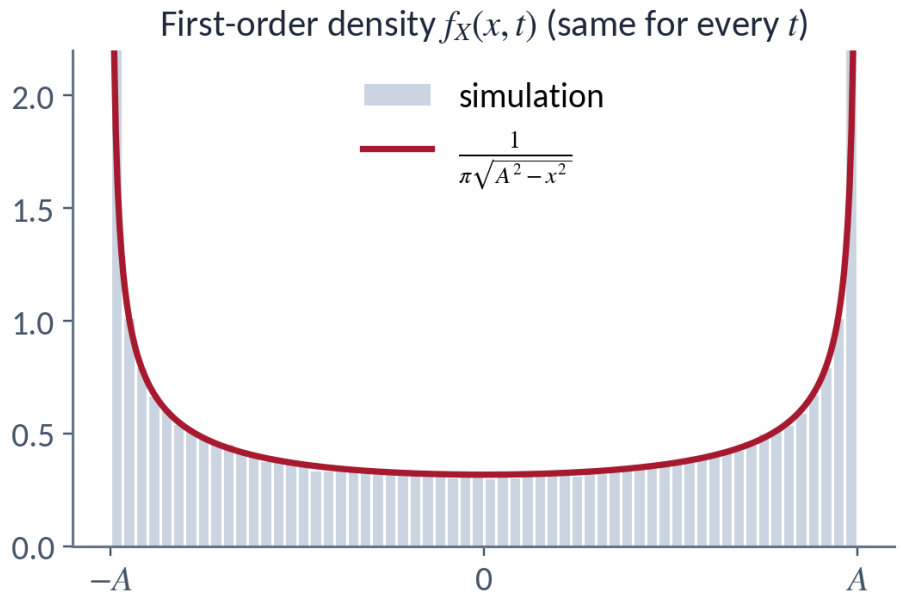
Example 2 (a): First-Order PDF

$A\cos(\omega_0 t + \theta_i) = x$ has exactly two solutions $\theta_1, \theta_2 \in (0, 2\pi]$ for $|x| < A$

$$\left|\frac{dX_t}{d\theta}\right|_{\theta_i} = \left|A\sin(\omega_0 t + \theta_i)\right| = \sqrt{A^2 - x^2}$$

RESULT: THE ARCSINE DENSITY

$$f_X(x,t) = \sum_{i=1}^{2} \frac{f_\theta(\theta_i)}{|dX_t/d\theta|_{\theta_i}} = \frac{2}{2\pi} \cdot \frac{1}{\sqrt{A^2 - x^2}} = \frac{1}{\pi\sqrt{A^2 - x^2}}, \quad |x| < A$$



Example 2 (b, c): Mean and Variance

(b) MEAN

$$E[X(t)] = \int_0^{2\pi} A\cos(\omega_0 t + \theta)\,\frac{1}{2\pi}\,d\theta = 0$$

(c) VARIANCE

$$\mathrm{Var}[X(t)] = E[X^2(t)] = \frac{A^2}{2\pi}\int_0^{2\pi}\frac{1+\cos(2\omega_0 t + 2\theta)}{2}\,d\theta = \frac{A^2}{2}$$

NOTE

Here **both** the mean and the variance are constant in t, unlike Example 1. In fact $E[X(t+\tau)X(t)] = (A^2/2)\cos(\omega_0\tau)$ depends only on τ: a first taste of **stationarity**.

Looking Ahead: Stationary Processes

Stationary Process

STRICT-SENSE STATIONARY (SSS)

$$f_X(x_1, \ldots, x_n;\, t_1 + \tau, \ldots, t_n + \tau) = f_X(x_1, \ldots, x_n;\, t_1, \ldots, t_n) \quad \forall\, n, \tau$$

All statistics are invariant to a time shift.

WIDE-SENSE STATIONARY (WSS)

$$E[X(t)] = \mu, \qquad E[X(t+\tau)X(t)] = R_X(\tau)$$

Only the mean and autocorrelation are shift-invariant. SSS ⇒ WSS (finite second moments).

Details next class

Practice Problems

1. For $X \sim \text{Poisson}(\lambda)$, show that $E[X] = \text{Var}[X] = \lambda$.

2. $X \sim U(0, 1)$. Find the PDF of $Y = -(1/\lambda) \ln X$. Which family is it?

3. X, Y i.i.d. $N(0, 1)$. Show that $U = X + Y$ and $V = X - Y$ are independent (use the Jacobian).

4. Use Chebyshev to find n so that the fraction of heads in n fair-coin tosses is within 0.05 of 1/2 with probability ≥ 0.95.

5. $X(t) = A \cos \omega_0 t + B \sin \omega_0 t$, with A, B i.i.d. $N(0, \sigma^2)$. Find $E[X(t)]$, $\text{Var}[X(t)]$ and $f_X(x, t)$.

NEXT WEEK

Stochastic Processes

Stationary Stochastic Processes

Have a good day!